



Ex 5.29 (未完)

$F(\theta, z)$ 关于第一个变量是凸的; $F'(\theta, z)$ 为 F 关于第一个变量次梯度

$$g_t = \frac{1}{r(z_t)} F'(\theta_{t-1}, z_t) \quad r(z_t) = \frac{d^2}{dp} r(z_t)$$

$\bar{F}(\theta) := \mathbb{E}_{z \sim p} [F(\theta, z)]$, 是实际损失函数

$$\mathbb{E}_q [g_t | \theta_{t-1}] = \int \frac{1}{r(z)} F'(\theta_{t-1}, z) d q(z) = \int F'(\theta_{t-1}, z) d p(z) \in \partial \bar{F}(\theta)$$

g_t 是 $z \sim q$ 采样时对于 $\bar{F}(\theta)$ 的一个无偏估计

$$\mathbb{E}_q [\frac{1}{r(z)} F'(\theta, z)] = \mathbb{E}_p [F'(\theta, z)] \quad \|F'(\theta, z)\|_2 \leq B(z) \quad \mathbb{E}_p B(z) \leq B$$

只要梯度有界 $\|g_t\| \leq \mathbb{E}_q \sup_z \frac{B(z)}{r(z)} = M(q)$

只要 $\|\theta_0 - \theta^*\| \leq D$, 梯度有界 B , 就可以证明收敛

$$\mathbb{E}_{z \sim q} \left[\frac{\|F'(\theta, z)\|^2}{(d^2/dp)^2} \right] \leq \mathbb{E}_{z \sim q} \frac{B^2(z)}{(d^2/dp)^2} = \frac{d^2}{dp} \mathbb{E}_{z \sim q} \frac{B^2(z)}{r(z)} = \frac{B^2}{r^2}$$

$$\begin{aligned} \mathbb{E}_{z \sim q} \|\theta_t - \theta^*\|^2 &= \mathbb{E}_{z \sim q} \|\theta_{t-1} - \gamma g_t - \theta^*\|^2 = \mathbb{E}_{z \sim q} \|\theta_{t-1} - \theta^*\|^2 - 2\mathbb{E}_{z \sim q} \gamma g_t^T (\theta_{t-1} - \theta^*) + \mathbb{E}_{z \sim q} \gamma^2 \|g_t\|^2 \\ &= \mathbb{E}_{z \sim q} \|\theta_{t-1} - \theta^*\|^2 - 2\mathbb{E}_{z \sim q} F'(\theta_{t-1})^T (\theta_{t-1} - \theta^*) + \mathbb{E}_{z \sim q} \gamma^2 \frac{B^2(z)}{r^2(z)} \end{aligned}$$

$$\mathbb{E}_{z \sim q} \sum_{s=1}^t \gamma_s F(\theta_s, z_s) - F(\theta^*) \leq \mathbb{E}_{z \sim q} \|\theta_0 - \theta^*\|^2 + \sum_{s=1}^t \gamma_s^2 \frac{B^2}{r^2}$$

控制 = 阶数 γ^2 $\therefore$ 令 $r(z) \propto B(z)$, 对 θ = 阶数重新估计

$$\text{考虑 SUM} \quad L(\theta) = \frac{1}{n} \sum_{i=1}^n \sum_{t=1}^T \log(1 + e^{-y_i x_i^T \theta})$$

$$\text{原本 } p \text{ 均采样} \quad g_t = \frac{1}{n \gamma_i} s_{it}(\theta_{t-1}) \quad i_t \sim q \quad s_{it}(\theta_{t-1}) = \begin{cases} -y_{it} x_{it} \\ -2y_{it} x_{it} \\ 0 \end{cases}$$

$$q_i \propto x_i \quad \|B(z)\| \propto \|x_i\| \quad q_i \propto \|x_i\|_2$$



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EX5.28 EX5.30 F 是 L -smooth, SGD 选择步长 γ , 使用无偏估计

EX5.29 梯度有上界 B^2

$$E\| \theta_t - \theta^* \|_2^2 \leq E\| \theta_{t-1} - \theta^* \|_2^2 - \gamma E\langle F'(\theta_{t-1}), \theta_t - \theta_{t-1} \rangle + \frac{\gamma^2}{2} E\| g_t(\theta_{t-1}) \|_2^2 \leq$$

$$F(\theta_t) \leq F(\theta_{t-1}) + \langle \nabla F(\theta_{t-1}), \theta_t - \theta_{t-1} \rangle + \frac{L}{2} \|\theta_t - \theta_{t-1}\|_2^2 \quad L\text{-smooth}$$

$$\theta_t = \theta_{t-1} - \gamma g_t(\theta_{t-1}) \quad E g_t(\theta_{t-1}) | \theta_{t-1} = \nabla F(\theta_{t-1})$$

$$E F(\theta_t) \leq E F(\theta_{t-1}) - \gamma E \| \nabla F(\theta_{t-1}) \|_2^2 + \frac{\gamma^2}{2} E \| g_t(\theta_{t-1}) \|_2^2 \leq$$

$$\text{中间是因为 } E \nabla F(\theta_{t-1})^T g_t(\theta_{t-1}) = E (E(\nabla F(\theta_{t-1})^T g_t(\theta_{t-1}) | \theta_{t-1})) = E \nabla F(\theta_{t-1})^T E g_t(\theta_{t-1}) \\ = E \| \nabla F(\theta_{t-1}) \|_2^2$$

但 $E \| g_t(\theta_{t-1}) \|_2^2$ 无法估计, 这是二阶矩了, 只能用通用上界 B^2

$$E F(\theta_t) \leq E F(\theta_{t-1}) - \gamma E \| \nabla F(\theta_{t-1}) \|_2^2 + \frac{\gamma^2}{2} B^2$$

$$E \| \nabla F(\theta_{t-1}) \|_2^2 \leq \frac{E F(\theta_0) - E F(\theta_t)}{\gamma} + \frac{LB^2\gamma}{2}$$

$$\frac{1}{t} \sum_{s=1}^t E \| \nabla F(\theta_{s-1}) \|_2^2 \leq \frac{E F(\theta_0) - E F(\theta_t)}{\gamma t} + \frac{LB^2\gamma}{2} \leq \frac{F(\theta_0) - \inf F(\mathcal{Z})}{\gamma t} + \frac{LB^2\gamma}{2}$$

Prop